

the results that are hoped to be answered or achieved.

set what we consider valid detection to generate bounding box. What is the accuracy of bottles and cups. Percentage. 50 percent cup. Confidence. How quick to detect cup or bottle. Time. Rates in codes. Change something? Or accuracy? Will it detect more the bottle and cup? How accurate? Can i add more objects? What is the metric? Is the robot able to detect a cup accurately in the right position? Why only cup and bottle?

The efficiency of the arm detection is out of the scope of this project. As in the future the arm should be working out of the box and just needs frames calibration. Solved by having specific place to be attached to. Fixed TF Interested in aesthetics as well as compact solution that is easy to set up. Efficiency of manipulation accuracy of manipulator how many times it actually grabbed? 60%

In order to know how accurate and effective this method can be in locating objects and grasping them.

dont need top view, dont need 2d camera, i get true location relative. design is compact, can have many applications. no view blocking.

Before using the Yolo method, the surface and object detection method was tested, and for this particular project this method didn't work as expected because of the noise in the image coming from the surrounding area, and the need to detect the surfaces of every object existing in the field of view then filtering those surfaces out to finally narrow down few objects to detect. For the point cloud, the surfaces detected needed to be filtered into surfaces of a certain height in which the objects will be on top. This method can take too much processing and filtering as well as it relies on color detection and has a limited amount of objects to detect from, compared to the Yolo method used in this project, Which is why this method was ruled out.

In figure (17) few screenshots show the number of surfaces that needed to be filtered in order to finally detect a banana and even though it was detected the camera view had too much noise making it unable to detect it accurately.

Figure (17): Detecting using this method requires a more controlled environment.

publish rate chatter

- accepted probability
- rospy, rate
- timeout python
- list size
- decrease list size

Will it detect more of the bottle and cup ?

- Is the robot able to detect a cup accurately in the right position ?
- How accurate?
- Can i add more objects ?
- Rates in codes. Change something ? Or accuracy ?
- How quick to detect cup or bottle. Time.

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